

This presentation can be found with audio at this unlisted YouTube URL:

<https://youtu.be/OMqgP50I7jo>

This PDF version is a place-marker for Moodle.

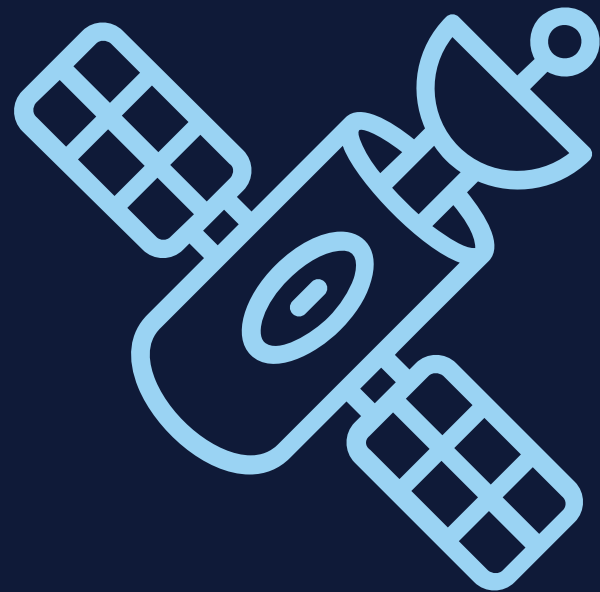
From Launch to Litter: Orbital Overload



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DAT5001_S1_25 PRES1



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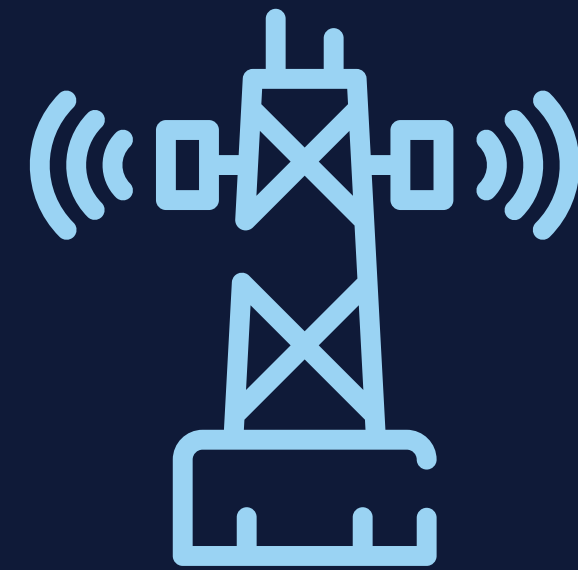
What is the Kessler Syndrome?

And its potential impact on civilization.



Exploring the Data

Looking at what's in orbit and who put it there.



Planning for The Future

How can we avoid a disaster?



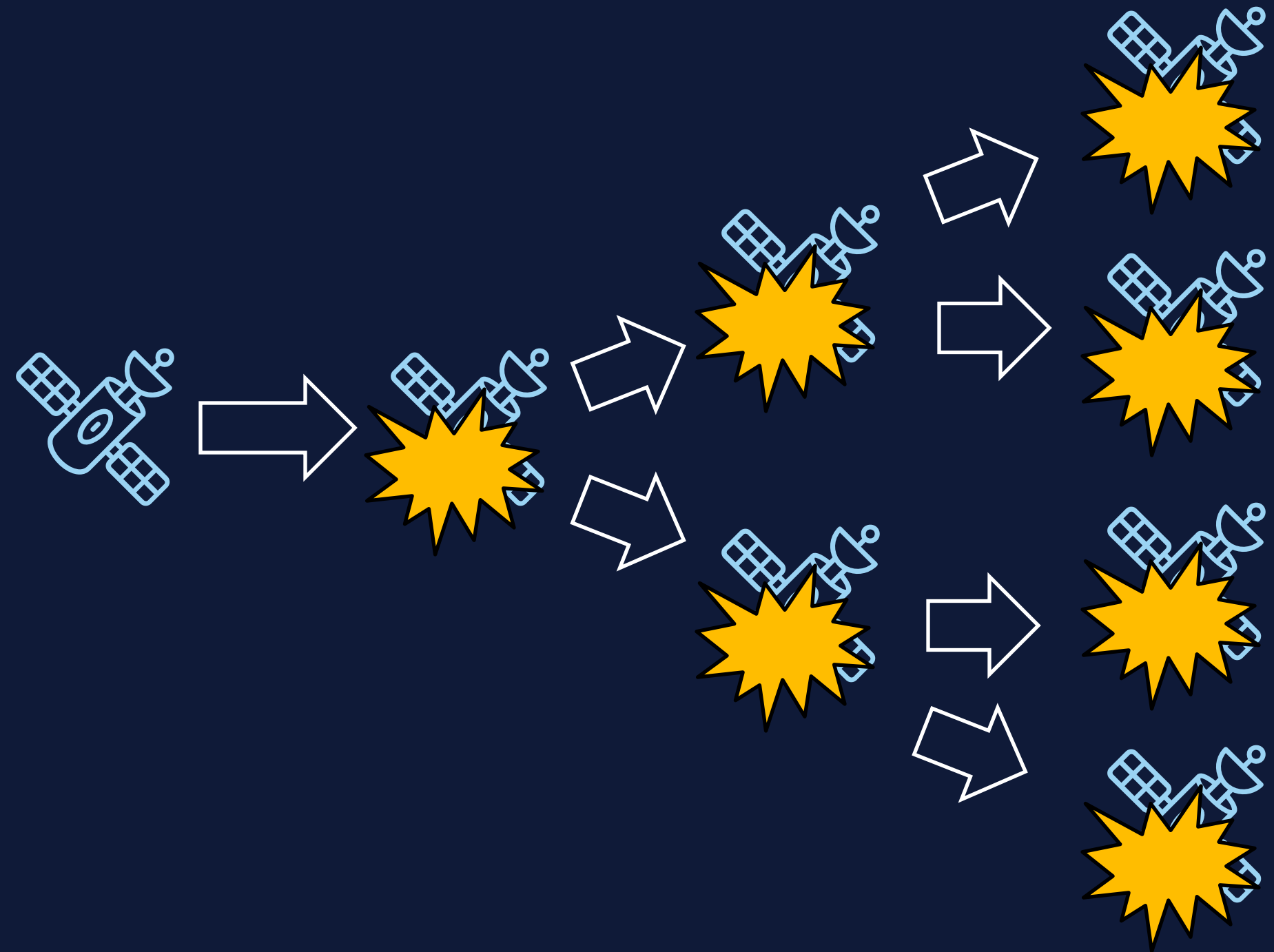
What is the **Kessler Syndrome**?

Collision Cascade

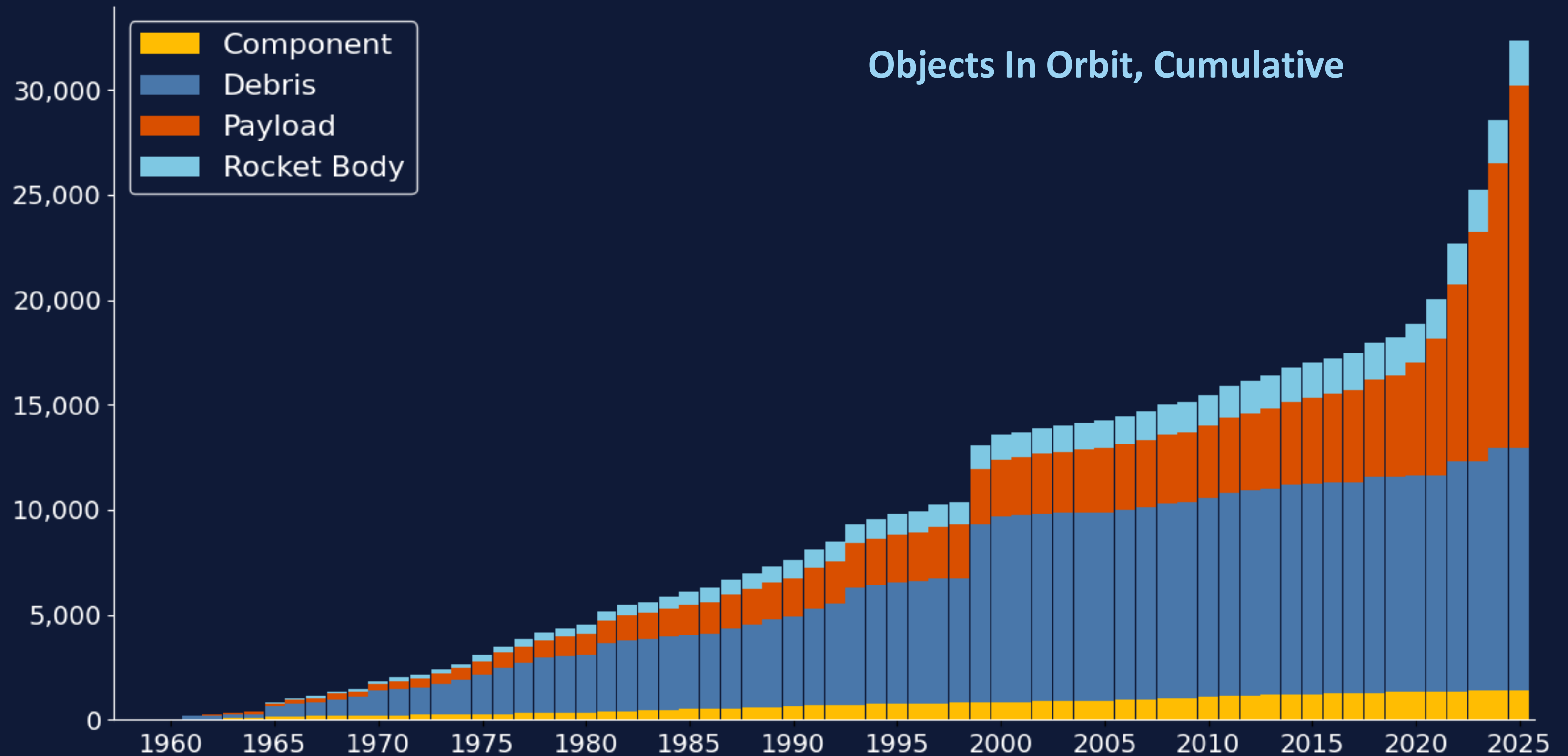
A scenario where satellite and/or space debris collisions cause a cascade effect, making low Earth orbit unusable.

Disaster Scenario

First identified in 1978 by Donald J. Kessler and Burton G. Cour-Palais.

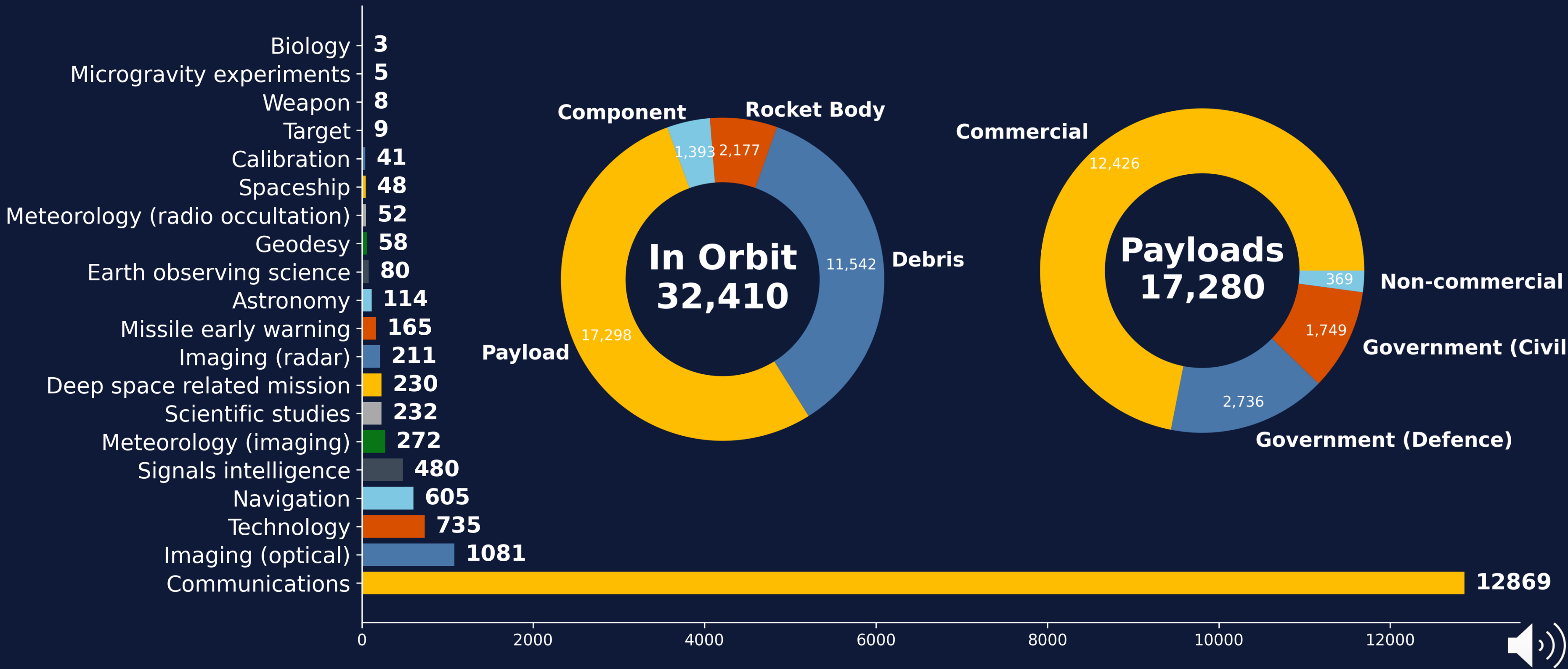


Satellite Launch Trends

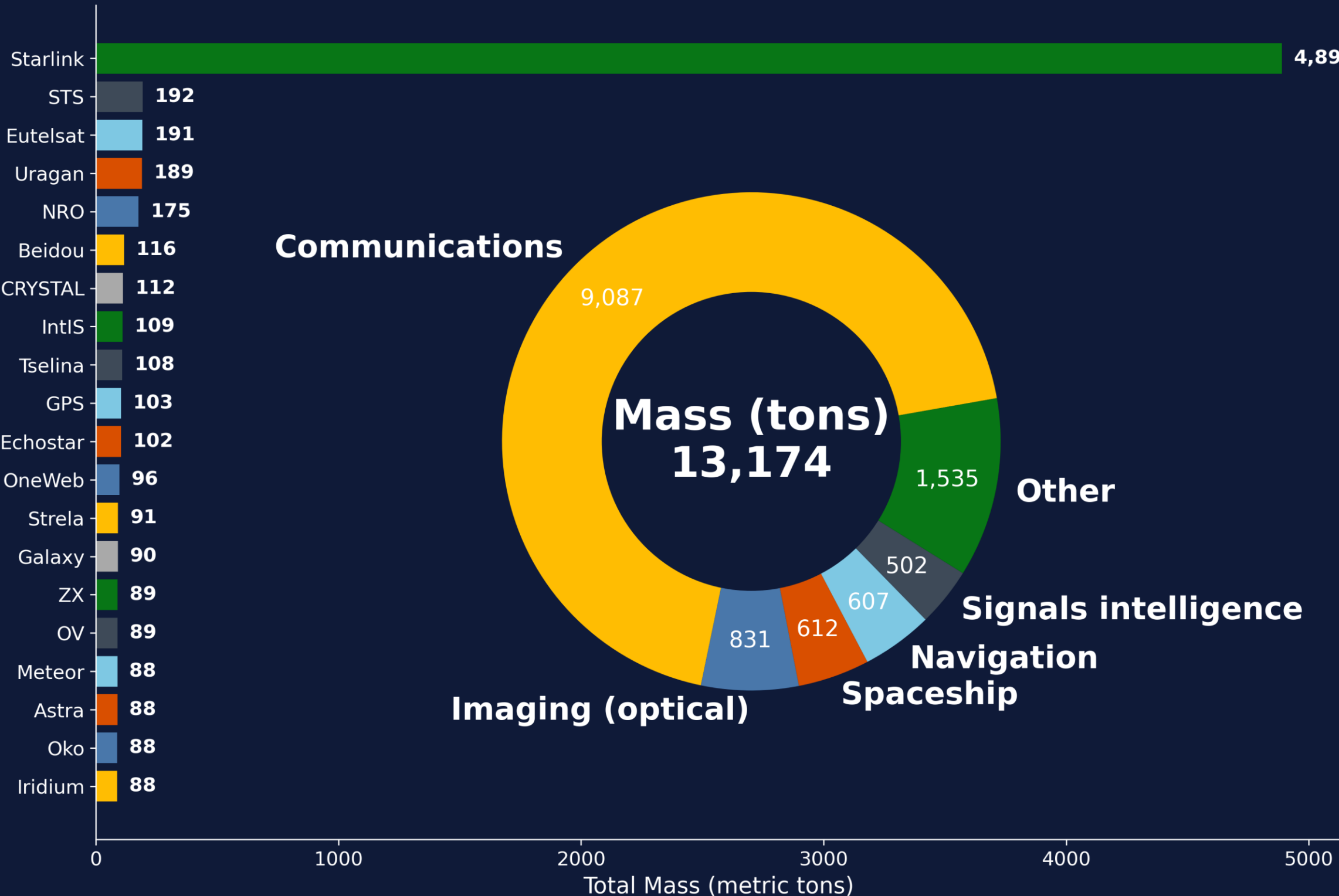


What's up in Orbit?

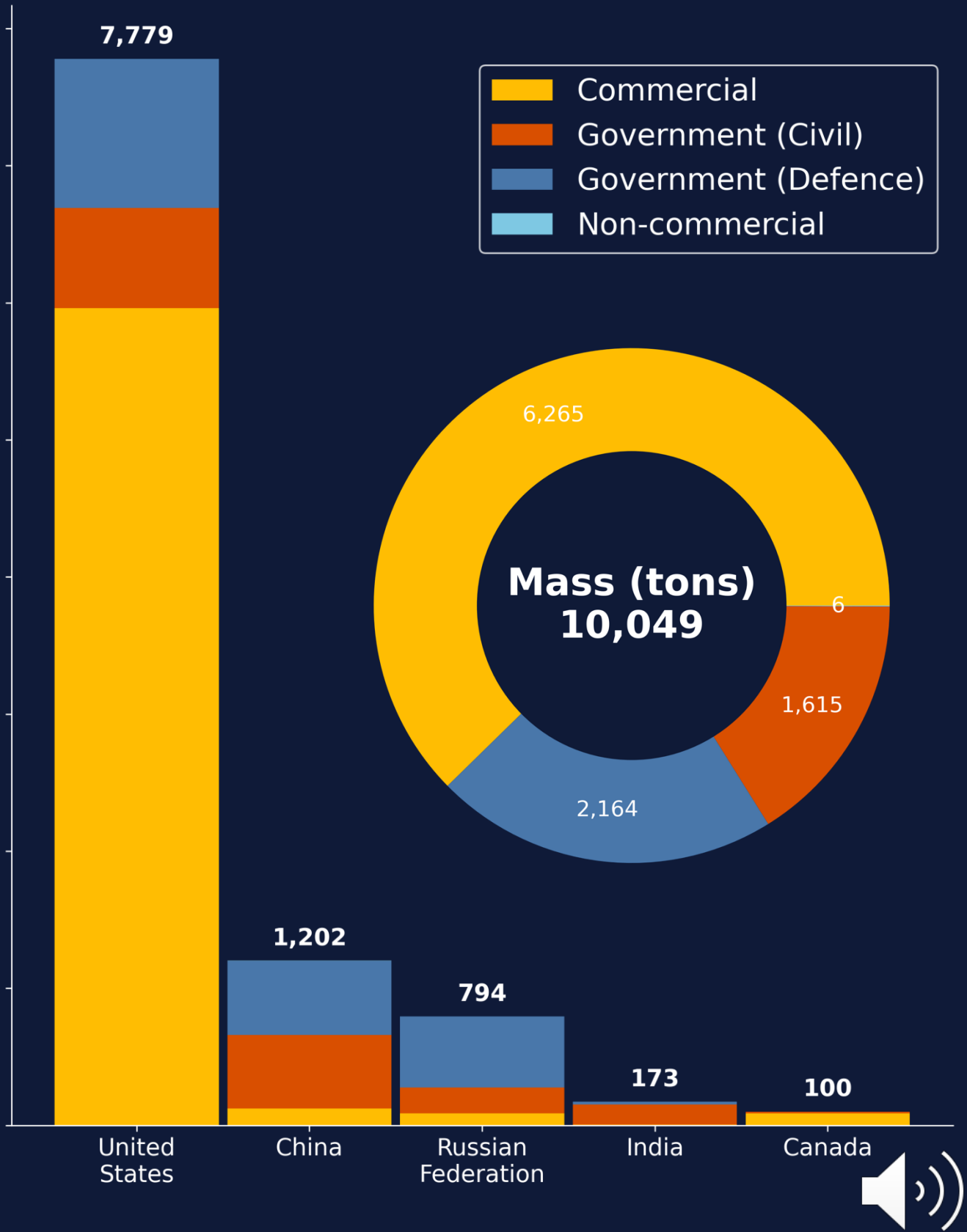
Count of Objects In Low Earth Orbit



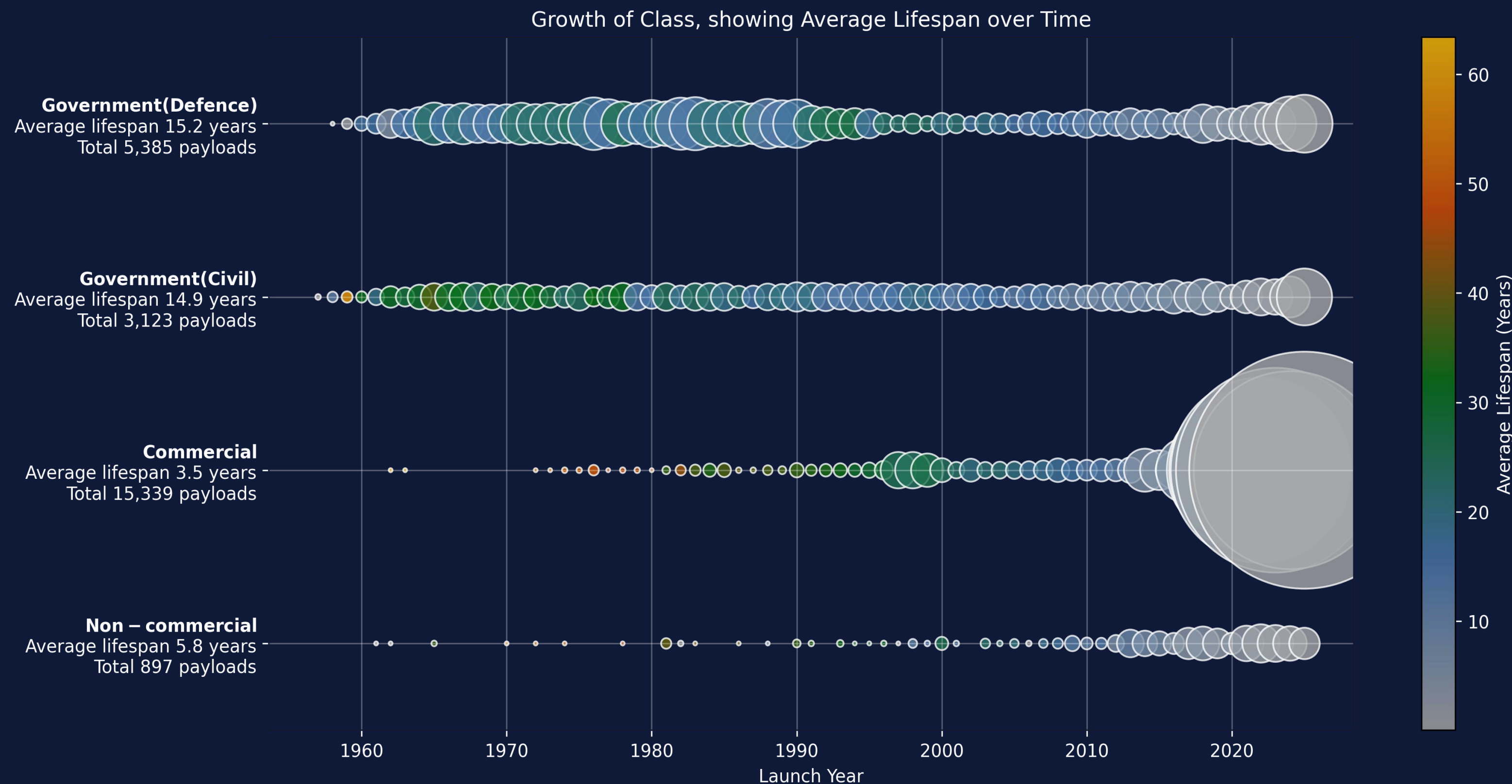
Heavy Metal



Mass of Payloads in Low Earth Orbit



Follow the Money



Operators and Sponsors

Starlink

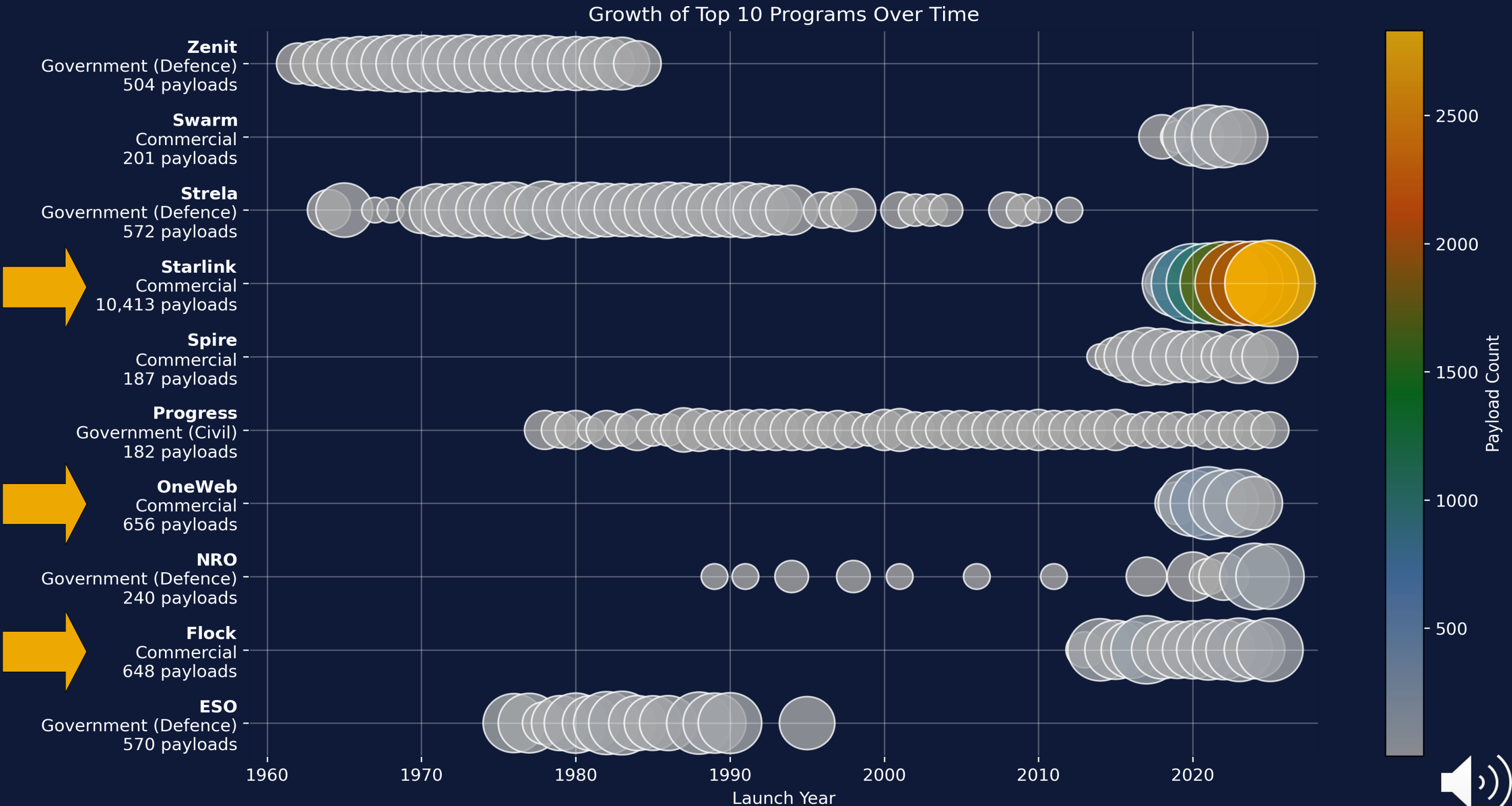
First launch: 2018
Total launches: 10,413
Purpose: Comms

OneWeb

First launch: 2019
Total launches: 656
Purpose: Internet

Flock

First launch: 2015
Total launches: 548
Purpose: Earth
Observation



Global Growth

1960

Total Countries: 2
Total Satellites: 46

1980

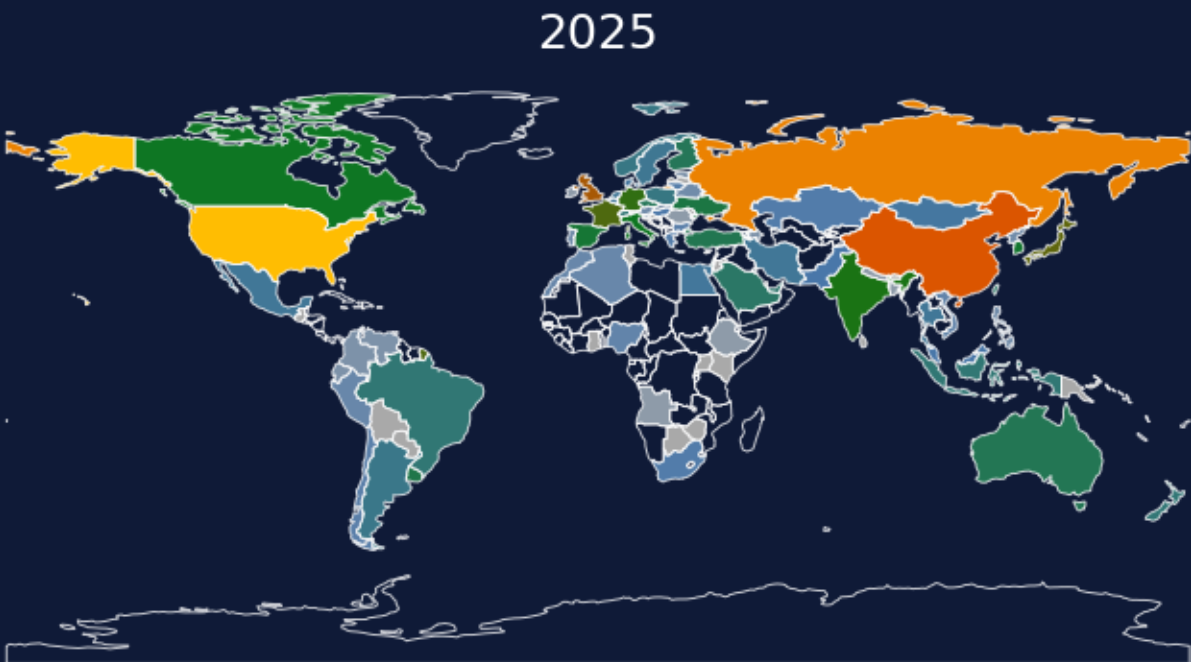
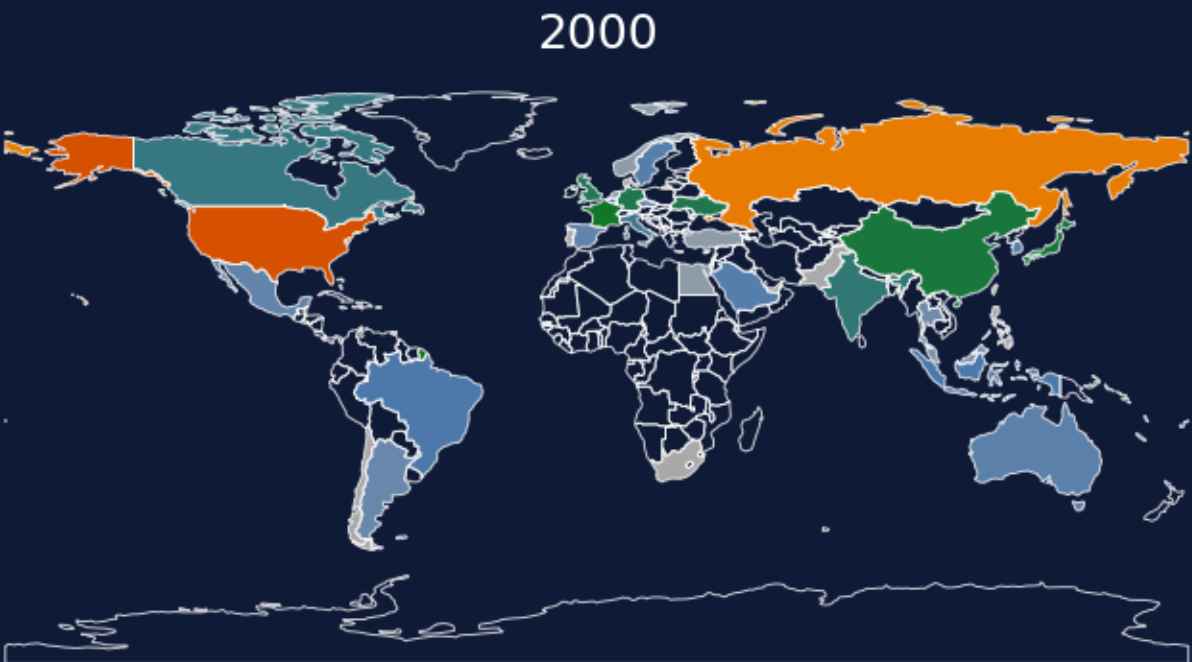
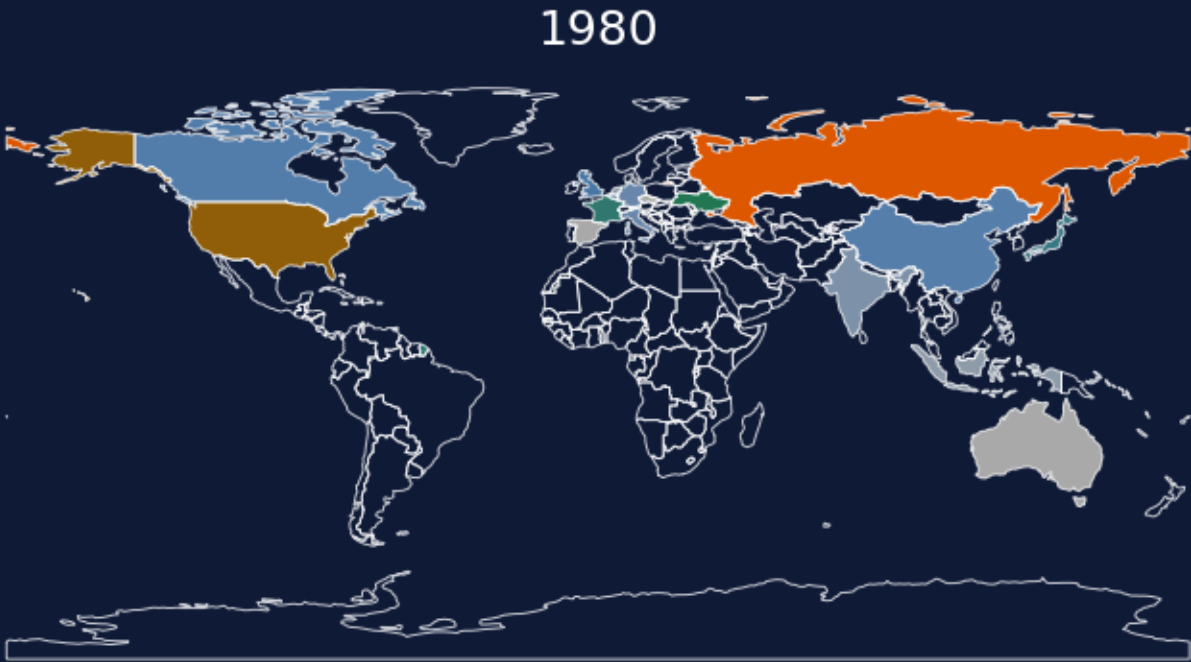
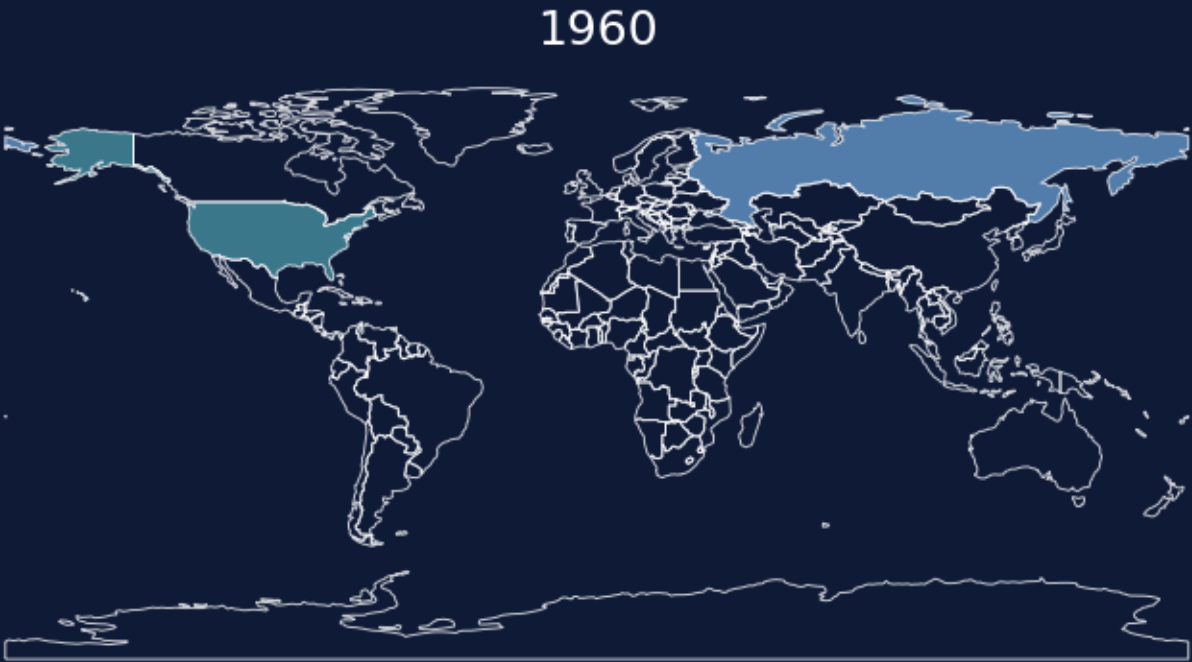
Total Countries: 17
Total Satellites: 2,834

2000

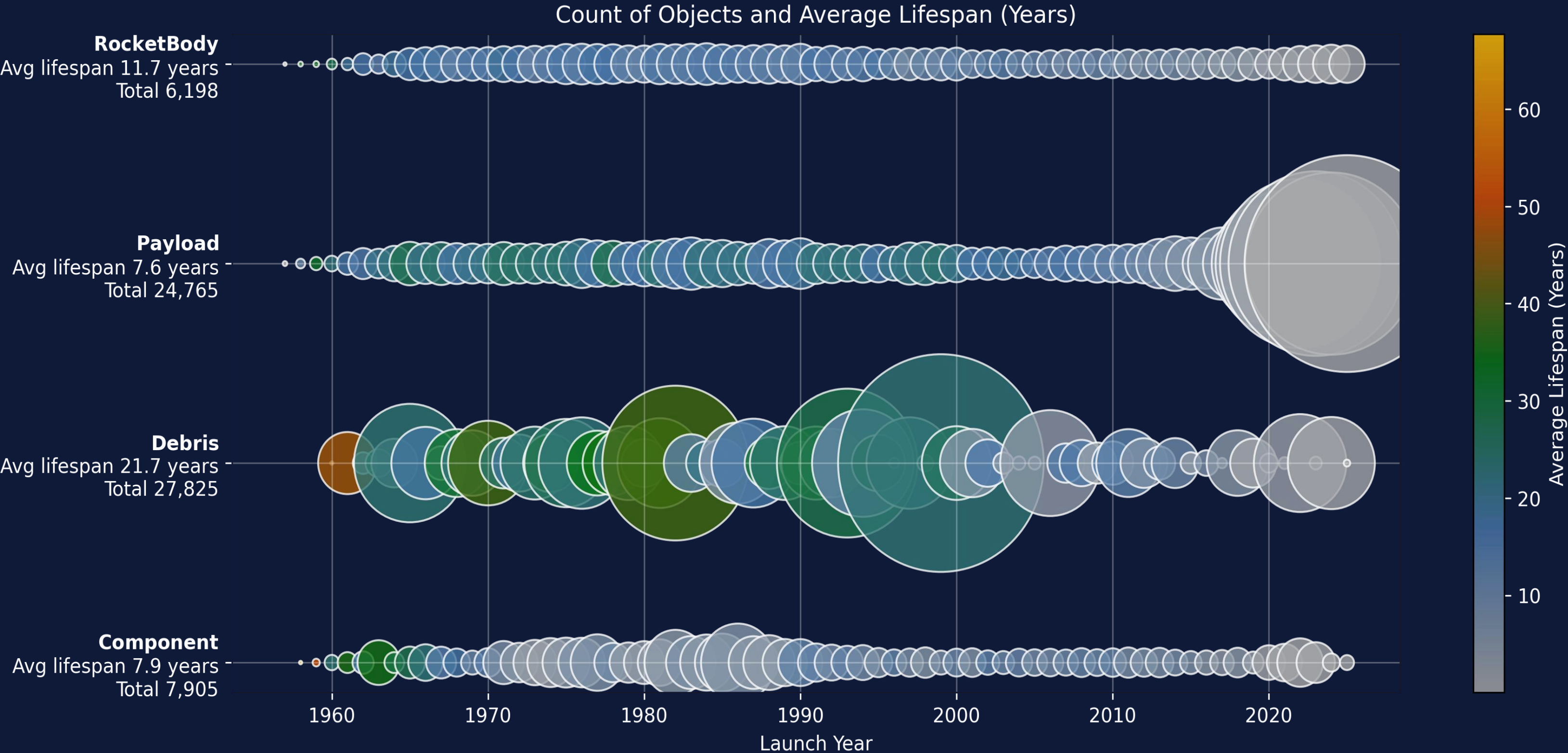
Total Countries: 39
Total Satellites: 6,026

2025

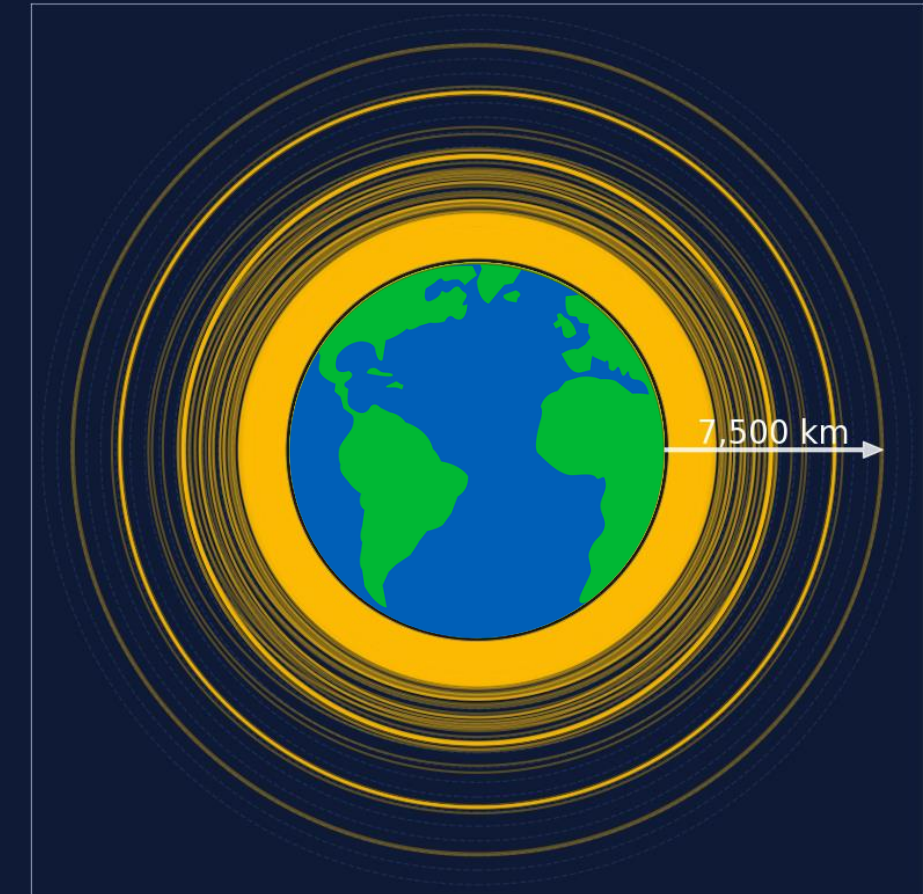
Total Countries: 98
Total Satellites: 24,756



Long-lived Litter



Where are **they** all?



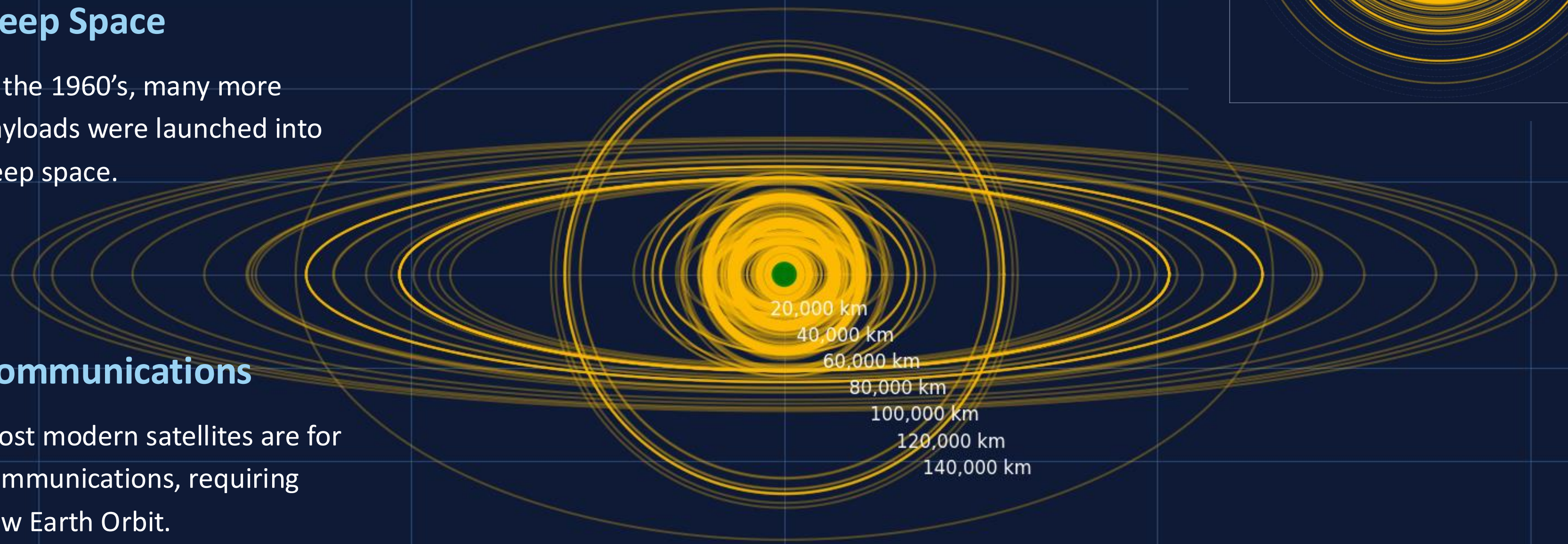
Orbital Paths of Payloads currently in Orbit

Deep Space

In the 1960's, many more payloads were launched into deep space.

Communications

Most modern satellites are for communications, requiring Low Earth Orbit.



What is **at Stake?**



A New Dark Age

Global Communications, GPS,
Internet, TV, Climate Change
Monitoring and Emergency
Response – **All Gone**

Economic Disaster

Space Industry is worth \$400B.

Knock on effects to all Earth
industry.

No More Missions

Space exploration would halt for
safety reasons.

ISS, Tiangong abandoned, Moon
and Mars missions cancelled.

Security and Defence

Loss of surveillance, weapons
detection and defence capability.



Stop it **and Tidy up**

Active Debris Removal

Capture and de-orbit defunct payloads.

Expensive.

Collision Avoidance

Automated systems to monitor orbits and adjust trajectory.

Design for De-orbit

Satellites designed to decay and burn up.

Cheaper than recovery, but new payloads only.

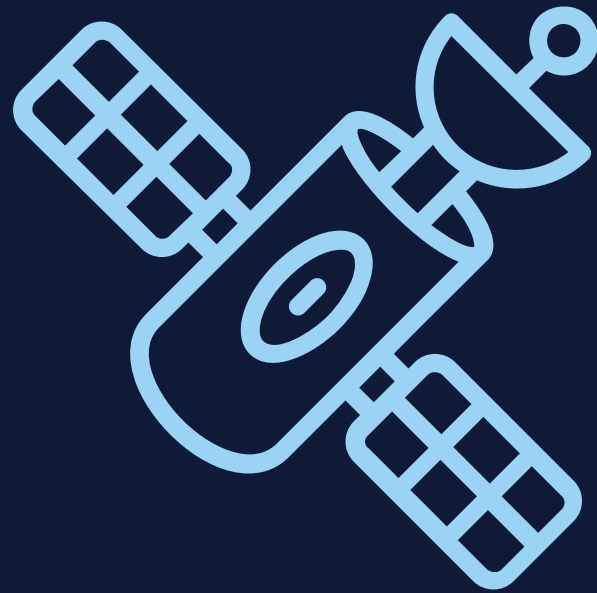
Policy and Agreements

International Guidelines (UN COPUOS).

Voluntary and weakly enforced.



In Conclusion



Devastating Kessler Cascade

A new dark age: loss of navigation, communications and more.



Low Earth Orbits are Crowded

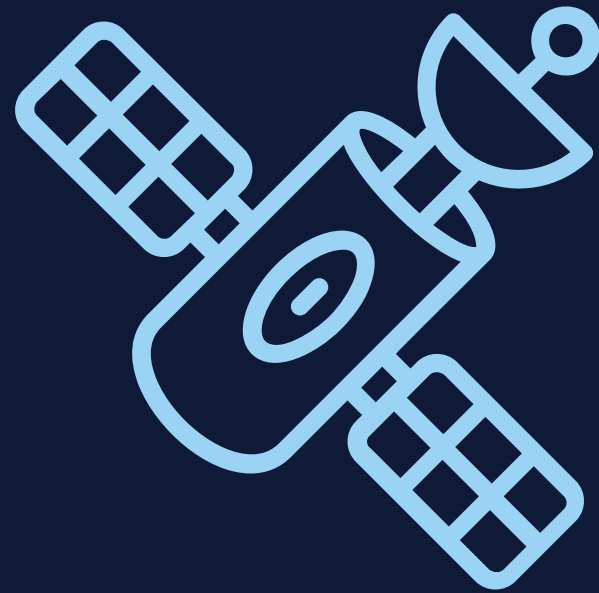
There is a lot of hardware in space, and we are adding to it at a record pace.



Planning for The Future

Without coordinated efforts, we will reach a tipping point within decades.





More Information:

<https://orbital.daemonchild.com>

Music: [Piano Dreamcloud Meditation, Natures Eye](#)

<https://pixabay.com/music/meditationspiritual-piano-dreamcloud-meditation-179215/>

